Tree Health Investigator

Ironhack DSMLFT September 2026

Project 3 AI Product

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Why build the Tree Health Investigator?

The problem:

- Increase in tree death and damage
- Increase in pest due to climate change

The challenges:

- Literature is complex
- Many different sources
- Symptoms can have multiple causes

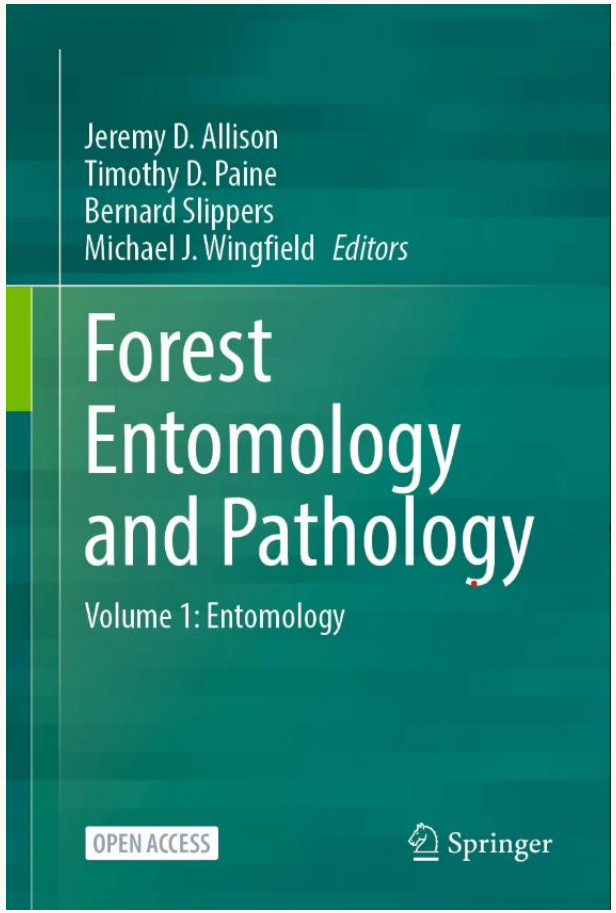
The Goal:

Help tree owners make better informed decisions.

Who is it for?

- Garden owners
- Hobby gardeners
- People with small forests

The data



- Allison, J. D., Paine, T. D., Slippers, B., & Wingfield, M. J. (Eds.). (2023). *Forest Entomology and Pathology: Volume 1: Entomology*. Springer.
- Open-access
- forest insects:
  - their ecological roles
  - different feeding groups
- insect-related tree damage
- approaches to forest pest management

Building and improving the AI model

Building

- Embedding model:
  - All-mpnet-base-v2, 768 dimensions
- Vector search:
  - ChromaDB, top-k=5
- Generation:
  - Gtp-6-luna
- Prompt design:
  - Evidence grounded (page citation)
  - Uncertainty handling

Improving

- Retrieval quality
  - Selecting MPNet
- Adding conversation history for follow-up questions
- Evaluated response time. Token usage and estimated cost

$0.27
1,000 queries

50 %
Retrieval performance

5.58 s
Average latency

1,795
tokens per query

Conclusion

What works well:

- 50 % retrieval success in testing
- 79,2 % correct answers
- Assistant produces relevant and evidence-based responses

Limitations:

- Limited to the book
- retrieval isn't always accurate
- It is rather slow

What I would do next:

- Expand the knowledge
  - Different pests and diseases
- Adding image-based symptom recognition

Do you have any questions?

Fell free to ask me later!



Project completed during the

Data Science and Machine Leaning Bootcamp

at **Ironhack**

Project duration: 16/10/2026-23/10/2026